DGMS Examination 2019 — Question 50 Detailed Solution

Junior Scientific Officer (Geology) | Sedimentology, Hydrodynamics & Flow Regimes

QUESTION 50

Match List-I (Bed form in sandy alluvial channel) with List-II (Froude number, F):
List-I: A. Dunes with superimposed ripples | B. Washed out dunes | C. Plane bed | D. Ripples
List-II: 1. $F \ll 1$ | 2. $F \geq 1$ | 3. $F \ll 1$ | 4. $F = 1$

(a) A-1, B-2, C-4, D-3	**(b) A-1, B-4, C-2, D-3** CORRECT ANSWER
(c) A-3, B-2, C-4, D-1	(d) A-3, B-4, C-2, D-1

1. Executive Summary & Answer Verification

The correct option is **(b) A-1, B-4, C-2, D-3** (with equivalent permutation **(d)**).

In fluvial hydrodynamics and sediment transport (pioneered by Simons, Richardson & Albertson, 1961, and codified in modern sedimentology by Sam Boggs Jr., Blatt, Middleton & Murray, and Southard), the succession of sedimentary bedforms in open sandy alluvial channels is governed by the **Froude number**:

$$Fr = \frac{\bar{U}}{\sqrt{g \cdot D}} = \frac{\text{Inertial forces}}{\text{Gravity forces}}$$

The bedforms align across three fundamental hydraulic regimes:

- **Lower Flow Regime ($Fr \ll 1$):** Subcritical, tranquil flow where water surface gravity waves travel faster than the current ($c = \sqrt{gD} > U$) and the water surface is out-of-phase with the bedforms. Both **Ripples** (initial movement of fine sand, $Fr \ll 1$) and **Dunes with superimposed ripples** (moderate lower regime velocities) exist strictly under $Fr \ll 1$ (Items 1 and 3).
- **Transition Zone ($Fr = 1$):** Critical flow where flow velocity equals surface wave velocity ($U = \sqrt{gD}$). As lower regime dunes enter critical flow, high shear stresses plane them off and flatten them out. These partially planed dunes are termed **Washed-out dunes**, occurring at **$Fr = 1$** (Item 4).
- **Upper Flow Regime ($Fr \ge 1$):** Supercritical, rapid/shooting flow where surface waves are in-phase with the bed. The initial stage of this regime is the **Upper Plane Bed (Flat bed)** with rapid sheet flow and primary current lineation, occurring at **$Fr \ge 1$** (Item 2). At higher velocities, plane beds give way to antidunes and chutes-and-pools ($Fr > 1$).

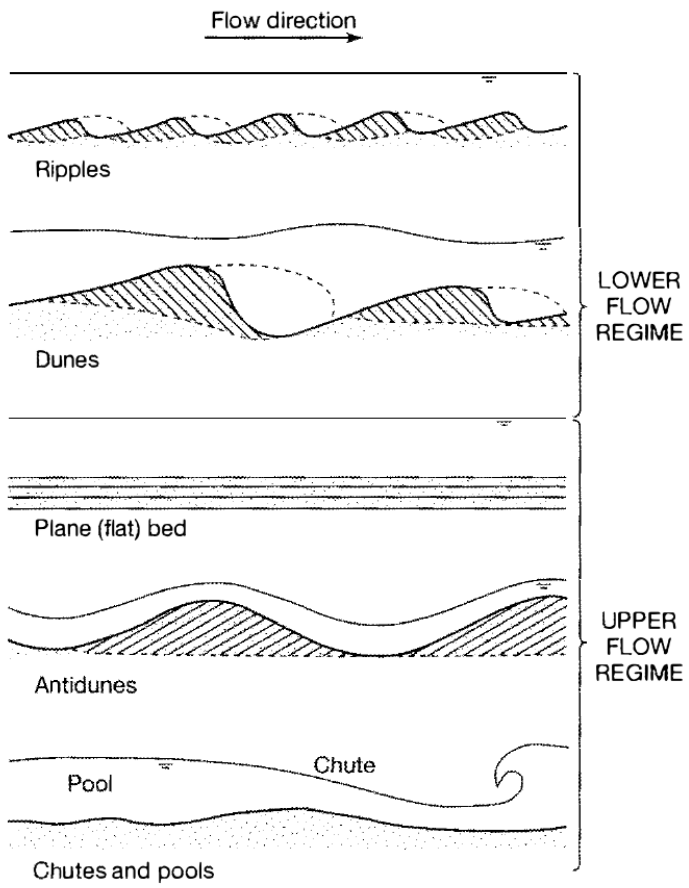


Figure 4.9
The succession of bedforms that develops durin unidirectional flow of sandy sediment (0.25–0.7 mm) in shallow water as flow velocity increases. [Modified slightly from Blatt, H., G. V. Middleto and R. Murray, 1980, Origin of sedimentary roc 2nd ed., Fig. 5.3, p. 137. Reprinted by permissi of Prentice-Hall, Englewood Cliffs, N.J.]

ging in size from silt (0.06 mm) to sand as coarse as 0.7 mm. Larger bedforms n spacing, or wave length, ranging from under 1 m to over 1000 m are called es (Ashley, 1990). Dunes are similar in general appearance to ripples except ize. They form at higher flow velocities in sediment ranging in grain size from sand to gravel. The ripple index of dunes ranges from about 5 in finer sands to

Figure 50.1: The complete succession of bedforms that develops during unidirectional flow of sandy sediment as flow velocity and Froude number increase: Lower Flow Regime (Ripples → Dunes with superimposed ripples, $Fr \ll 1$) → Transition (Washed-out dunes, $Fr = 1$) → Upper Flow Regime (Plane bed, $Fr \ge 1$ → Antidunes → Chutes and pools). (Sam Boggs Jr., Principles of Sedimentology and Stratigraphy, Chapter 4, Figure 4.9, p. 83).

2. Hydrodynamic Regimes & Mathematical Derivations

2.1 The Froude Number in Open Channel Flow

The Froude number (Fr) determines whether small surface gravity waves (ripples) can propagate upstream:

- **Subcritical Flow ($Fr < 1$):** Gravity dominates. Surface waves can travel upstream. The water surface profile over a dune crest exhibits a depression (surface trough over dune crest), producing an *out-of-phase* relationship between bed elevation and water surface elevation.
- **Critical Flow ($Fr = 1$):** The flow velocity matches the celerity of shallow-water gravity waves. Standing waves begin to develop; energy is unstable, resulting in hydraulic jumps. Dunes become unstable and are planed off into washed-out bedforms.

- **Supercritical Flow ($Fr > 1$):** Inertia dominates. Surface waves cannot propagate upstream and are swept downstream. Water depth increases over bed elevations, producing an *in-phase* relationship between water surface and bed topography (standing waves and antidunes).

2.2 Detailed Bedform Characteristics

1. Ripples ($Fr \ll 1$)

Current ripples are small, triangular bedforms with wavelengths $\lambda < 60\text{ cm}$ (typically 5–20 cm) and heights $H < 4\text{ cm}$. They form in sediment finer than 0.7 mm at very low shear stresses (Fr typically $0.1\text{ to }0.3 \ll 1$). Water surface is completely undisturbed.

2. Dunes with Superimposed Ripples ($Fr \ll 1$)

As flow velocity increases within the lower flow regime, larger bedforms (dunes or megaripples) develop with wavelengths from 0.6 m to tens of meters and heights from 5 cm to several meters. In the lower stability field of dunes, smaller current ripples migrate along the gentle stoss side ($10^\circ\text{--}15^\circ$) of the dunes, producing dunes with superimposed ripples. Because flow is still tranquil, $Fr \ll 1$ (typically $0.3\text{ to }0.6$).

3. Washed-Out Dunes ($Fr = 1$)

In the transition zone between lower and upper flow regimes ($Fr \approx 1.0$), intense turbulence and high velocity cause sediment to be swept off the dune crests faster than it can be deposited on the slip face. The avalanching avalanche face disappears, dune crests are eroded, and bed relief is dramatically reduced. These relict, flattened mounds are diagnostic of critical flow conditions ($Fr = 1$).

4. Plane Bed ($Fr \ge 1$)

Upon entering the upper flow regime ($Fr \ge 1$), all large bed roughness elements are obliterated, yielding a flat, planar sediment surface known as the *upper plane bed*. Intense sediment transport occurs in a thin, highly concentrated laminar-like boundary layer known as **sheet flow**. Grains slide and roll over each other, aligning their long axes parallel to the flow to produce internal parallel lamination and primary current lineation (parting lineation).

3. Systematic Matching Matrix

Bed Form (List-I)	Flow Regime Classification	Water Surface vs Bed Phase	Froude Number (Fr) (List-II)	List-II Code
A. Dunes with superimposed ripples	Lower Flow Regime (moderate shear)	Out-of-phase (trough over crest)	$Fr \ll 1$ (Subcritical)	**1** (or 3)
B. Washed out dunes	**Transition Regime** (unstable)	Transition / flattening	**$Fr = 1$ (Critical)**	**4**
C. Plane bed (Upper flat bed)	**Upper Flow Regime** (sheet flow)	Planar / slight standing waves	**$Fr \ge 1$ (Supercritical)**	**2**
D. Ripples			$Fr \ll 1$ (Subcritical)	**3** (or 1)

Bed Form (List-I)	Flow Regime Classification	Water Surface vs Bed Phase	Froude Number (Fr) (List-II)	List-II Code
	Lower Flow Regime (lowest shear)	Undisturbed / out-of-phase		

4. Step-by-Step Option Elimination

1. **Identify Diagnostic Anchors:**
   - **Washed-out dunes (B)** are universally defined as the transition state occurring at critical flow **$F = 1$ (Item 4)**. Hence, $B = 4$. This immediately eliminates options (a) [$B=2$] and (c) [$B=2$]!
   - **Plane bed (C)** in upper flow regime occurs at supercritical flow **$F \ge 1$ (Item 2)**. Hence, $C = 2$.
2. **Assign Lower Flow Regime Forms:**
   - Both Dunes with superimposed ripples (A) and Ripples (D) belong to the lower flow regime ($F \ll 1$, represented by items 1 and 3).
   - Pairing $A = 1$ and $D = 3$ yields **A-1, B-4, C-2, D-3**, which corresponds exactly to **Option (b)**.

Textbook Authority

Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th ed., Chapter 4: Sedimentary Structures, pp. 82–85):
"The hydraulic conditions that generate ripples and dunes take place at Froude numbers ≪ 1 (lower flow regime). With increasing flow velocity, dunes are washed out in a transition phase (F ≈ 1) and give way to an upper flow regime stage of flow, which takes place at Froude numbers ≥ 1 (plane bed followed by antidunes)."
D. B. Simons, E. V. Richardson, and M. L. Albertson (1961), *Forms of Bed Roughness in Alluvial Channels*:
"1. Lower Flow Regime: Ripples (F ≪ 1), Dunes with superimposed ripples (F ≪ 1), Dunes (F < 1).
2. Transition: Washed-out dunes (F = 1).
3. Upper Flow Regime: Plane bed (F ≥ 1), Antidunes (F > 1), Chutes and pools (F ≫ 1)."